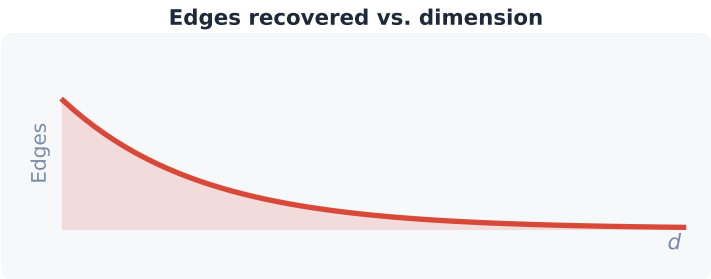
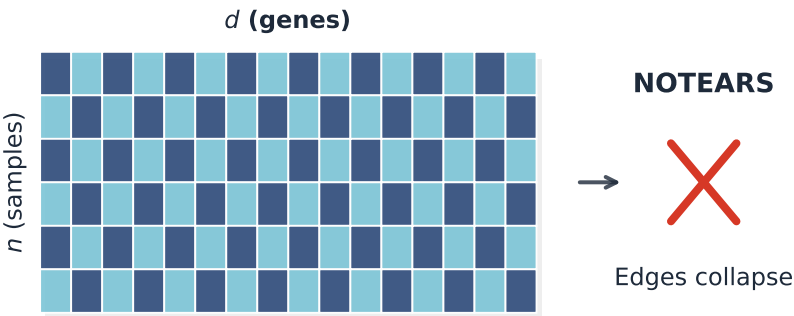


CAGate: Cluster-Aware Gating for Differentiable Causal Discovery in Cancer Genomics

Rescuing high-dimensional causal structure learning through residual clustering and MAD-based gating

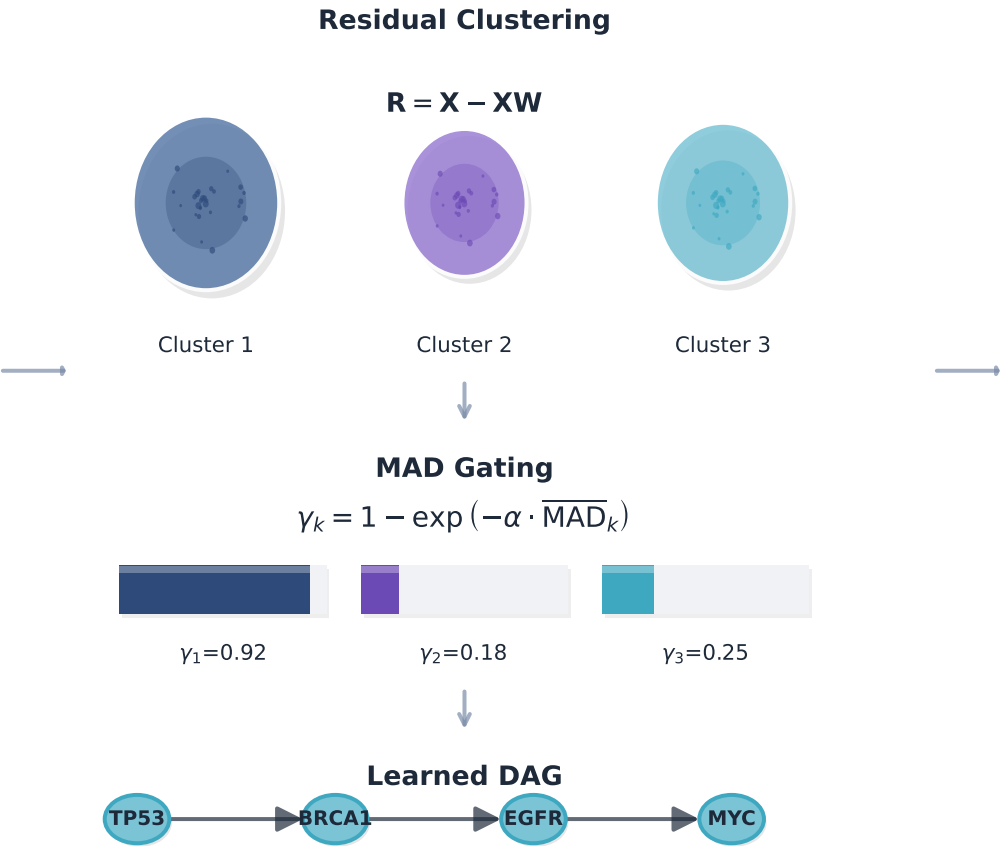
High-dimensional failure

$$n \ll d$$

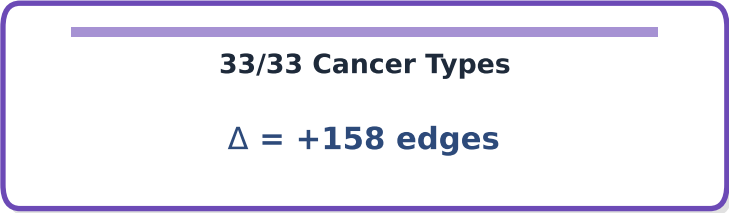


When $n \ll d$, traditional DAG methods fail to recover meaningful edges

CAGate: Cluster-Aware Gating



Pan-cancer rescue

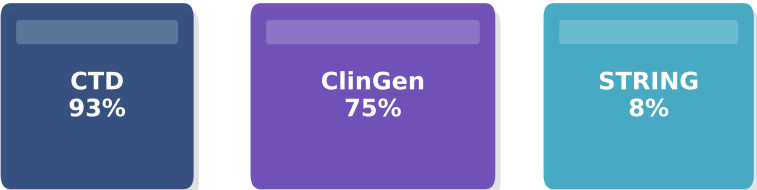


High-dimensional recovery



collapses in high dimensions

External validation



Small-sample advantage

+237 ($n < 308$) vs +84 ($n \geq 308$)